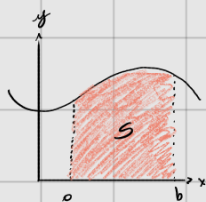


Relembrando que

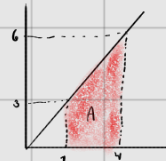
* Teorema fundamental do cálculo: $\int_a^b f(x) dx = F(b) - F(a)$



$$S = \int_a^b f(x) dx$$

$$* \int_1^4 (x+2) = \left. \frac{x^2}{2} + 2x \right|_1^4 = \left(\frac{4^2}{2} + 2 \cdot 4 \right) - \left(\frac{1^2}{2} + 2 \right) = \frac{27}{2}$$

$$f(x) = x+2 \rightarrow f(x) \geq \forall x \in [1, 4]$$

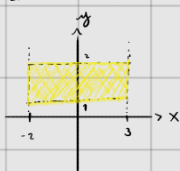


INTEGRAL DUPLA:

↳ O integrando é uma função de duas variáveis

↳ Esboçar regiões para a cálculo do integral re termo recuário.

a) $[-2, 3] \times [1, 2]$



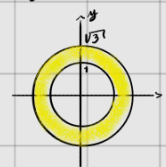
b) $x^2 + y^2 \leq 4$



c) região formada por $y=2$, $y=x+2$ e $x=4$



d) $x^2 + y^2 = 1$ e $x^2 + y^2 = 3$



e) $y = x^2$ e $y = x$



* Uma não esboçar de região.

Calcular os integrais:

$$a) \int_1^2 \int_0^1 \frac{x}{y} dx dy = \int_1^2 \left[\frac{x^2}{2} \right]_0^1 dy = \int_1^2 \frac{1}{2} dy = \frac{1}{2} \left[y \right]_1^2 = \frac{1}{2} (\ln 2 - \ln 1) = \frac{1}{2} \ln 2 \text{ ou } \ln 2^{\frac{1}{2}} \text{ ou } \ln \sqrt{2} //$$

$$b) \int_1^2 \int_0^{\frac{\pi}{2}} x \cos(y) dy dx = \int_1^2 \left[x \sin(y) \right]_0^{\frac{\pi}{2}} dx = \int_1^2 x \cdot 1 dx = \left[\frac{x^2}{2} \right]_1^2 = \frac{2^2}{2} - \frac{1^2}{2} = \frac{3}{2} //$$

$$c) \int_0^1 \int_{x^2}^x (x+y) dy dx = \int_0^1 \left[\int_{x^2}^x x dy + \int_{x^2}^x y dy \right] dx = \int_0^1 \left[\left. x y \right|_{x^2}^x + \left. \frac{y^2}{2} \right|_{x^2}^x \right] dx = \int_0^1 \left[x(x - x^2) + \left(\frac{x^2}{2} - \frac{x^4}{2} \right) \right] dx = \int_0^1 \left[-\frac{x^4}{2} - x^3 + \frac{3}{2} x^2 \right] dx = \textcircled{I}$$

$$\textcircled{I} \left[-\frac{x^5}{10} - \frac{x^4}{4} + \frac{3}{2} x^3 \right]_0^1 = -\frac{1}{10} - \frac{1}{4} + \frac{3}{2} = \frac{3}{20} //$$

* Fazer em Cor: $\int_1^2 \int_0^1 y e^{x+1} dx dy = \frac{3e-1}{2}$

$$y \cdot e^1 - y e^0 = y \cdot e - y = y \cdot (e-1)$$

$$\int_1^2 \int_0^1 y e^{x+1} dx dy = \int_1^2 \left[\int_0^1 y e^x dx + \int_0^1 1 dx \right] dy = \int_1^2 \left[y \cdot e^x \Big|_0^1 + x \Big|_0^1 \right] dy = \int_1^2 y \cdot (e-1) + 1 dy =$$

$$= \int_1^2 y \cdot (e-1) + 1 dy = \int_1^2 y \cdot (e-1) dy + \int_1^2 1 dy = (e-1) \cdot \frac{y^2}{2} \Big|_1^2 + y \Big|_1^2 = (e-1) \cdot \left(\frac{4}{2} - \frac{1}{2} \right) + (2-1) = \frac{3}{2} (e-1) + 1 = \frac{3e-3+2}{2} = \frac{3e-1}{2} //$$

* Agora com duas variáveis, vamos fazer varreduras na região de integração:

1) A região de integração é um retângulo e/ou lados paralelos aos eixos coordenados.



R pode ser descrito como:

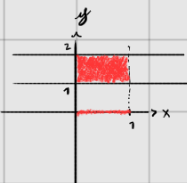
$\hookrightarrow$ Região delimitada por $x=a$, $x=b$, $y=c$, $y=d$
ou
 $\hookrightarrow [a, b] \times [c, d]$

$$\rightarrow \int_a^b \int_c^d f(x, y) dy dx \text{ ou } \int_c^d \int_a^b f(x, y) dx dy$$

EXEMPLO:

①

$$R = [0, 1] \times [1, 2]$$



$$\iint_R 3xy^2 dA \Rightarrow \int_0^1 \int_1^2 3xy^2 dy dx \text{ ou } \int_1^2 \int_0^1 3xy^2 dx dy \quad * \text{ Para escolher o que ser mais conveniente}$$

$$* \int_1^2 \int_0^1 3xy^2 dx dy = \int_1^2 3y^2 \cdot \left(\frac{x^2}{2} \Big|_0^1 \right) dy = \int_1^2 3y^2 \cdot \frac{1}{2} dy = \int_1^2 \frac{3}{2} y^2 dy = \frac{3}{2} \cdot \left(\frac{y^3}{3} \Big|_1^2 \right) = \frac{3}{2} \cdot \frac{7}{3} = \frac{7}{2} //$$

② $R = [0, \pi] \times [0, 1]$



$$\iint_R x \sin(xy) dA = \int_0^\pi \int_0^1 x \sin(xy) dy dx \text{ ou } \int_0^1 \int_0^\pi x \sin(xy) dx dy$$

$$\int_0^\pi \int_0^1 x \sin(xy) dy dx = \int_0^\pi 1 - \cos x dx = \int_0^\pi 1 dx - \int_0^\pi \cos x = x \Big|_0^\pi - (\sin x \Big|_0^\pi) = \pi - (-1+1) = \pi$$

$$* \int_0^1 x \sin(xy) dy = x \cdot \int_0^1 \sin(xy) dy = x \cdot \int_0^1 \sin(u) \cdot \frac{du}{x} = \int_0^1 \sin(u) du = -\cos(xy) \Big|_0^1 = -\cos(x) - (-\cos 0) = -\cos x + 1 = 1 - \cos x$$

$u = xy \quad \frac{du}{dy} = x$
 $dy = \frac{du}{x}$

$$\iint_R x \, dA$$

* Simples B limitada pelos curvos $y = 5x - x^2$ e $y = 2x$



* Pontos em que as funções se encontram

$$\begin{aligned} 5x - x^2 &= 2x \\ x^2 - 3 &= 0 \\ x &= 0 \\ x &= 3 \end{aligned}$$

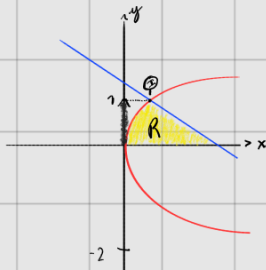
$$R = \begin{cases} 0 \leq x \leq 3 \\ 2x \leq y \leq 5x - x^2 \end{cases}$$

* Em x é a variação que vamos ter para o integral.

$$\int_0^3 \int_{2x}^{5x-x^2} x \, dy \, dx = \int_0^3 x y \Big|_{2x}^{5x-x^2} dx = \int_0^3 x (5x - x^2 - 2x) dx = \int_0^3 (-x^3 + 3x^2) dx = \left[-\frac{x^4}{4} + x^3 \right]_0^3 = -\frac{81}{4} + 27 = \frac{54}{4}$$

Região em função de y ($x = \text{sentença de } y$)

Calcular o integral duplo $\iint_R (y+1) \, dA$, A é a região delimitada por $y^2 = x$, $y = -x+2$ e $y=0$.



$$\textcircled{2} \begin{cases} y = -x+2 \rightarrow y = -y^2+2 \\ x = y^2 \end{cases}$$

$$y=1 \text{ ou } y=-2$$

$$R = \begin{cases} 0 \leq y \leq 1 \\ y^2 \leq x \leq 2-y \end{cases}$$

Entrar no R pelo parábola
Sair do R pelo reto

$$\int_0^1 \int_{y^2}^{2-y} (y+1) \, dx \, dy = \int_0^1 (y+1) x \Big|_{y^2}^{2-y} dy = \int_0^1 [y(2-y) + (2-y)] - [y \cdot y^2 + y^2] dy = \int_0^1 (-y^3 - 2y^2 + y + 2) dy = \left[-\frac{y^4}{4} - \frac{2}{3}y^3 + \frac{y^2}{2} + 2y \right]_0^1 = \frac{19}{12}$$

Calcular $\iint_R \frac{-y}{x^2} \, dA$, R é a região formada por $y=x$, $y=1$, $y=0$ e $x=-y^2+4$



$$R = \begin{cases} 0 \leq y \leq 1 \\ y \leq x \leq -y^2+4 \end{cases}$$

$$\rightarrow \int_0^1 \int_y^{-y^2+4} \frac{-y}{x^2} \, dx \, dy$$

$$\int_0^1 \int_y^{-y^2+4} \frac{-y}{x^2} \, dx \, dy = \int_0^1 \left[\frac{y}{-y^2+4} - 1 \right] dy = \int_0^1 \frac{y}{-y^2+4} dy - \int_0^1 1 \, dy = -\frac{1}{2} \ln|-y^2+4| \Big|_0^1 - y \Big|_0^1 = \left(-\frac{1}{2} \ln(3) \right) - \left(-\frac{1}{2} \ln(4) \right) - 1 = \frac{1}{2} \ln(4) - \frac{1}{2} \ln(3) - 1 = \frac{1}{2} \ln\left(\frac{4}{3}\right) - 1$$

$$\textcircled{I} \int_y^{-y^2+4} \frac{-y}{x^2} \, dx = \int_y^{-y^2+4} -y \cdot x^{-2} \, dx = -y \int_y^{-y^2+4} x^{-2} \, dx = -y \left[\frac{x^{-1}}{-1} \right]_y^{-y^2+4} = \frac{y}{x} \Big|_y^{-y^2+4} = \frac{y}{-y^2+4} - \frac{y}{y} = \frac{y}{-y^2+4} - 1$$

$$\textcircled{II} \int_0^1 \frac{y}{-y^2+4} dy = \int_0^1 \frac{y}{u} \frac{du}{-2y} = \int_0^1 \frac{1}{-2} \frac{1}{u} du = -\frac{1}{2} \int_0^1 \frac{1}{u} du = -\frac{1}{2} \ln u$$

$$u = -y^2+4$$

$$\frac{du}{dy} = -2y \rightarrow dy = \frac{du}{-2y}$$

$$\textcircled{III} \int_0^1 1 \, dy = y$$